

Market Concentration and Consumer Welfare in the Illinois Group Health Insurance Market: Evidence from Complaint Ratios and Loss Ratios

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Abstract

The Illinois group comprehensive health insurance market is one of the most concentrated in the United States: a single insurer, Health Care Service Corporation (HCSC, operating as Blue Cross Blue Shield of Illinois), holds approximately 79.66% of the market by direct premiums written, producing a Herfindahl–Hirschman Index of approximately 6,521, more than 2.6 times the U.S. Department of Justice and Federal Trade Commission threshold for a highly concentrated market. This paper asks whether that concentration translates into measurable consumer harm. Using publicly available 2024 data from the Illinois Department of Insurance Health Market Share Report and Consumer Complaint Ratio Report, supplemented by Kaiser Family Foundation reporting on medical loss ratios and Medicare Advantage prior authorization, the analysis constructs market shares, complaint ratios, complaint-composition shares, and pure direct loss ratios for the four major insurers active in the Illinois group comprehensive segment: HCSC, UnitedHealthcare, Aetna, and Health Alliance. Three principal findings emerge. First, HCSC generates the lowest consumer complaint ratio of any major insurer at 1.36 complaints per 10,000 policies, despite its dominant market position. Second, claims handling accounts for 53 to 79 percent of all complaints across every insurer regardless of size, indicating that adjudication practices, not market power, are the primary source of consumer friction in this market. Third, HCSC’s pure direct loss ratio of 92.24% exceeds the Affordable Care Act’s 85% medical loss ratio floor for large group plans, while two smaller competitors fall marginally below it. Taken together, these results imply that policy attention to consumer welfare in this market is better directed at claims-process transparency and accountability than at market structure alone.

Keywords: health insurance; market concentration; Herfindahl–Hirschman Index; consumer complaints; medical loss ratio; dominant-firm model.

JEL: I11, I13, L11, G22.

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Plain-Language Summary

One company, Blue Cross Blue Shield of Illinois (HCSC), sells roughly 80% of employer health plans in the state, making this one of the most concentrated insurance markets in the country. Economic theory usually warns that this kind of dominance hurts customers. This paper checks whether it actually does, using public state insurance data for 2024.

The result is counterintuitive. The dominant insurer draws the *fewest* complaints per customer, not the most. And regardless of a company's size, the same thing generates most complaints: how claims are handled, approvals, denials, and the paperwork around them, rather than market power itself.

On the money side, the dominant insurer pays out a high share of premiums as actual care, above the legal floor, so it does not look like it is extracting monopoly profits through price.

The practical takeaway: regulators worried about consumers here should focus less on breaking up market share and more on how claims get decided, denial rates, prior authorization, and appeals. The honest limit is that complaints undercount real denials, and this is a single year of data, so it shows strong patterns rather than proven cause and effect.

1 INTRODUCTION

Health insurance markets occupy a unique position in the study of industrial organization. Unlike most consumer markets, health insurance involves simultaneous failures across multiple dimensions, asymmetric information between insurer and policyholder, moral hazard on the demand side, significant barriers to entry on the supply side, and purchasing intermediation through employers that severs the direct link between consumer preference and market outcomes. These conditions make health insurance markets fertile ground for the kinds of strategic behavior and welfare distortions that market organization theory predicts in imperfectly competitive settings. Yet the empirical relationship between market concentration and consumer harm in health insurance remains contested, in part because the mechanisms linking structure to conduct to outcomes are more complex than standard oligopoly models suggest.

The Illinois group health insurance market offers a particularly instructive case. By any conventional measure, it is among the most concentrated commercial health insurance markets in the United States. A single insurer, Health Care Service Corporation, operating as Blue Cross Blue Shield of Illinois, controls approximately 79.66% of the group comprehensive market by premiums written, producing a Herfindahl–Hirschman Index of approximately 6,521. This figure is more than 2.6 times the DOJ/FTC threshold for a highly concentrated market and places Illinois closer to a monopoly than to a competitive market structure on the standard HHI spectrum. Against this backdrop, the central question this paper investigates is whether that concentration translates into measurable consumer harm.

The answer the data provides is more nuanced than the hypothesis predicts. Using publicly available data from the Illinois Department of Insurance, this paper constructs complaint ratios, complaint composition measures, and loss ratios for the four major insurers active in the Illinois group comprehensive market, HCSC, UnitedHealthcare, Aetna, and Health Alliance, and finds three principal results. First, HCSC generates the lowest consumer complaint ratio of any major insurer at 1.36 complaints per 10,000 policies, despite its dominant market position. Second, claims handling accounts for 53 to 79 percent of all complaints across every insurer, regardless of size, suggesting that adjudication practices rather than market power are the primary source of consumer friction. Third, loss ratios present a nuanced picture of regulatory effectiveness, the two largest insurers exceed the ACA's 85% MLR floor, while the two smaller competitors fall marginally below it, suggesting the regulatory constraint operates unevenly across firms of different scale.

These findings contribute to the market organization literature on dominant-firm markets and regulatory design in two ways. They suggest that HCSC's market dominance reflects demand-side employer preferences and accumulated network advantages rather than anticompetitive conduct producing measurable consumer harm by standard metrics. And they identify claims adjudication, driven by the asymmetric information structure inherent to insurance contracting and the moral hazard management incentives facing all insurers regardless of size, as the more direct source of consumer welfare concern in this market. This has implications for how policymakers prioritize interventions: antitrust enforcement targeting market structure may be less effective at improving consumer outcomes than regulatory reforms targeting claims transparency, prior authorization practices, and denial rate disclosure.

The remainder of the paper proceeds as follows. Section 2 details data sources, market definition, and the construction of market-share, complaint, and loss-ratio measures. Section 3 presents the empirical findings: market structure (Figures 1 and 2), the complaint-ratio puzzle (Table 1), the complaint-composition mechanism (Table 2), and the loss-ratio evidence (Table 3). Section 4 synthesizes these findings and presents policy implications and directions for future research.

2 METHODOLOGY

2.1 Data Sources

This paper draws on three primary data sources, all publicly available from government agencies and established health policy research organizations. Market share data is sourced from the Illinois DOI (2025b) 2024 Health Market Share Report, which compiles annual health insurance filings submitted by licensed carriers under the National Association of Insurance Commissioners (NAIC) reporting framework. Complaint data is sourced from the Illinois DOI (2025a) 2024 Consumer Complaint Ratio Report, which aggregates complaints filed directly with the Illinois Department of Insurance (DOI) by policyholders against licensed health insurers. The medical loss ratio context is drawn from the KFF (2024) 2024 Medical Loss Ratio Rebates analysis, which synthesizes insurer-reported financial data compiled by Mark Farrah Associates from NAIC filings. Prior authorization context is drawn from the KFF (2026) report on Medicare Advantage prior authorization determinations, based on CMS Part C public use data for contract years 2019–2024.

2.2 Market Definition and Sample

The analysis focuses exclusively on the group comprehensive line of health insurance business in Illinois for the 2024 reporting year. This segment covers employer-sponsored group health plans, the largest and most economically significant segment of the commercial health insurance market, and excludes individual comprehensive, Medicare Supplement, Medicaid managed care, dental-only, vision-only, and Medicare Advantage lines. This restriction is deliberate: the research question concerns the commercial employer-sponsored market, where HCSC's dominance is most pronounced and where prior authorization and claims adjudication mechanisms are most directly relevant to working-age consumers and their physicians.

2.3 Market Concentration Measures

Market share for each firm is calculated as the firm's Illinois direct premiums written (DPW) in the group comprehensive line divided by total Illinois DPW for that line, expressed as a percentage. The Herfindahl–Hirschman Index (HHI) is calculated as the sum of squared market share percentages across all firms active in the market, consistent with the standard DOJ and FTC (2023) merger guideline methodology. Firms are aggregated to the parent-company level where applicable, HCSC subsidiaries are consolidated under a single HCSC entry, UnitedHealth Group subsidiaries are consolidated, and Aetna subsidiaries are consolidated, to reflect competitive dynamics at the level of strategic decision-making rather than legal entity structure.

2.4 Complaint Ratio Construction

Complaint ratios are constructed as the number of complaints filed against a firm divided by the firm's policies in force as of December 31, 2024, multiplied by 10,000. This normalizes complaint counts by policyholder exposure, allowing meaningful comparison across firms of dramatically different sizes. Raw complaint counts and policies-in-force figures are drawn directly from the Illinois DOI (2025a) 2024 Complaint Ratio Report. As with market share, complaint figures are aggregated to the parent-company level by summing complaints and policies across all subsidiary entities reported separately in the IDOI data. For UnitedHealth, this aggregates UnitedHealthcare Insurance Company, UnitedHealthcare Insurance Company of Illinois, and UnitedHealthcare of Illinois Inc. For Aetna, this aggregates Aetna Life Insurance Company, Aetna Health Inc., and Aetna Health Insurance Company.

Complaint composition is measured as the share of each firm's total complaints attributable to each major reason category, claims handling, underwriting, marketing and sales, and policyholder service, as reported directly by the IDOI. Note that a single complaint may be assigned multiple reason categories, so reason-category counts may exceed total complaint counts. Claims handling share is therefore calculated as claims handling complaints divided by total complaints, treating multiple-reason complaints as fully attributable to each assigned category.

2.5 Loss Ratio Construction

Pure direct loss ratios are calculated as Illinois direct losses incurred divided by Illinois direct premiums earned for the group comprehensive line, expressed as a percentage. Both figures are drawn from the Illinois DOI (2025b) 2024 Health Market Share Report. This measure reflects the share of earned premium revenue paid out as incurred claims and is compared against the ACA's Medical Loss Ratio (MLR) floor of 85% applicable to large group insurers.

2.6 Limitations

Several limitations of this analysis warrant explicit acknowledgment. First, the complaint ratio measures reported consumer dissatisfaction rather than actual denial rates or adverse coverage decisions, and almost certainly undercounts the true frequency of such decisions. Second, loss ratios measure aggregate premium-to-claims flows but do not reveal the composition of approved versus denied claims within that aggregate. Third, prior authorization data from the KFF Medicare Advantage report is used for contextual framing only and is not directly applicable to the commercial group market analyzed here, Illinois-specific commercial prior authorization data is not publicly available at the insurer level. Fourth, the analysis is cross-sectional for a single year and cannot establish causal relationships between market structure and consumer outcomes. Fifth, aggregating UnitedHealth and Aetna subsidiaries introduces some imprecision, as policies-in-force figures for subsidiary entities reflect that entity's individual enrollment rather than a fully deduplicated parent-level count.

3 RESULTS AND DISCUSSION

3.1 Market Structure

The Illinois group comprehensive health insurance market is dominated by a single firm. Health Care Service Corporation (HCSC), operating as Blue Cross Blue Shield of Illinois, holds approximately 79.66% of the market by direct premiums written. UnitedHealthcare follows at 10.66%, Health Alliance at 7.73%, and Aetna at just 1.10%, with all remaining carriers accounting for less than 1% combined (Figure 1).

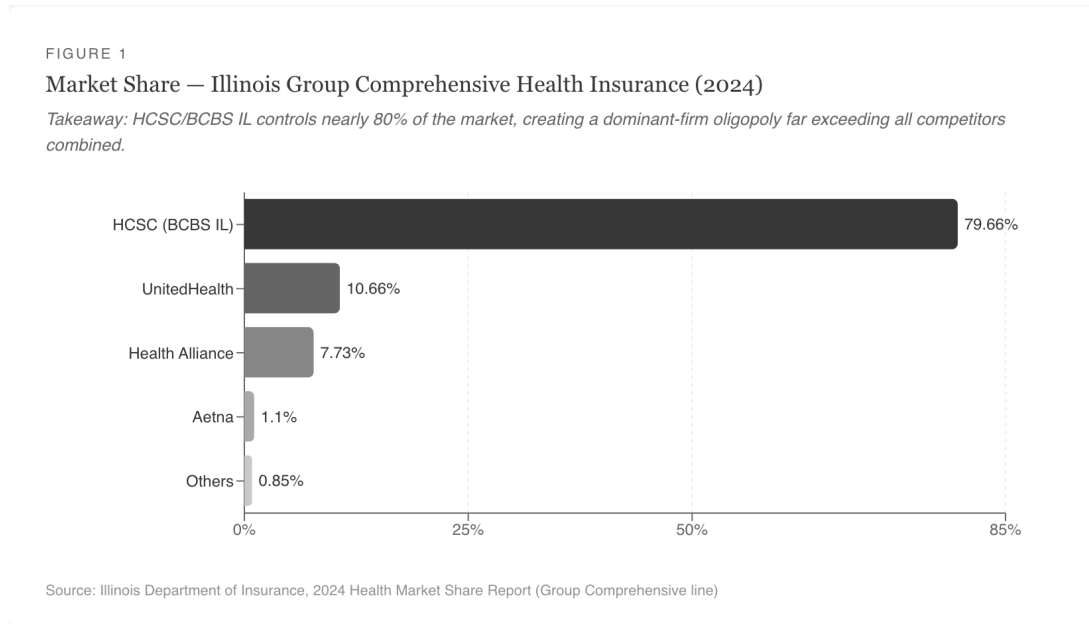


Figure 1: Market share by direct premiums written, Illinois group comprehensive health insurance, 2024. HCSC/BCBS IL controls nearly 80% of the market, creating a dominant-firm structure that exceeds all competitors combined. Source: Illinois DOI (2025b).

The structure of the Illinois group health insurance market reflects a classic dominant-firm model, one in which a single insurer commands a share so large that the remaining firms collectively constitute a competitive fringe rather than meaningful rivals. This matters because market structure is the foundational condition from which insurer behavior and consumer outcomes flow. When one firm controls nearly four out of every five insured group members in a state, the standard assumptions of competitive pricing and consumer choice break down.

The concentration here is not incidental. HCSC's dominance in the group comprehensive segment reflects decades of embedded relationships with Illinois employers, who serve as the actual purchasing decision-makers in employer-sponsored insurance markets. Unlike individual markets where consumers choose their own plans, the group market intermediates choice through employers, meaning BCBS's network breadth, administrative infrastructure, and brand familiarity with Illinois employers likely reinforce its position independent of price competition alone. This demand-side stickiness functions as a structural barrier to entry, limiting the ability of competitors like UnitedHealth or Aetna to meaningfully contest market share even if they offer comparable products.

Premium pricing in the group health insurance market operates through a mechanism that further insulates HCSC from conventional competitive pressure. Unlike consumer-facing markets where buyers observe and compare prices directly, group health premiums are set through annual bilateral negotiations between insurers and employers using experience-based actuarial pricing, that is, premiums are calibrated to the employer group's own historical claims experience rather than to a community-wide rate. This structure means that individual employees, who are the ultimate consumers of the insurance product, have no direct visibility into premium pricing and no meaningful ability to shop on price at enrollment. The competitive pressure that does exist operates at the employer level, through the annual contract renewal cycle, and is mediated by the administrative switching costs discussed above. The ACA's Medical Loss Ratio floor of 85% for large group plans functions as a regulatory constraint on this pricing dynamic, capping the share of premium revenue that insurers may retain above incurred claims and thereby limiting the extent to which dominant-firm pricing power can translate into extractive margins. As the loss ratio evidence in Section 3.5 demonstrates, this constraint appears to be binding in the Illinois group market, though unevenly across firms of different scales.

The strategic interaction structure of this market departs meaningfully from the standard oligopoly models, Cournot quantity competition or Bertrand price competition, that dominate introductory market organization analysis. In those frameworks, firms set quantities or prices simultaneously, anticipating rivals' responses, and equilibrium is reached through direct competitive interdependence. In the Illinois group health market, however, the primary competitive arena is the annual employer contract, not a continuous price-quantity market. HCSC's dominant position means it functions less as a strategic rival to UnitedHealth or Aetna in the classical sense and more as a price-and-service standard against which fringe competitors compete for residual demand, the dominant-firm-competitive-fringe model formalized by Stigler (1957) and extended in subsequent market organization literature. The more consequential strategic interaction in this market may be vertical rather than horizontal: the relationship between insurer and employer involves repeated bargaining over contract terms, network scope, and administrative service levels, with employer switching threats serving as the primary disciplinary mechanism. This employer-insurer vertical relationship, rather than horizontal rivalry among insurers, is the primary locus of competitive discipline in the market, and it helps explain why HCSC's complaint performance is superior despite its dominant position.

A key limitation of market share data alone is that it measures dominance but not conduct. A firm can hold 80% of a market and still behave competitively if structural conditions constrain its pricing power, or it can exploit that position to extract rents. Whether HCSC's dominance translates into consumer harm or operational efficiency is a question the subsequent figures and tables address directly.

3.2 Herfindahl-Hirschman Index

The HHI for the Illinois group comprehensive health insurance market is approximately 6,521, calculated as the sum of squared market shares across all firms (Figure 2). HCSC's share alone contributes 6,345.72 points to this total, with UnitedHealth contributing 113.64, Health Alliance 59.75, and Aetna and others accounting for less than 2 points combined.

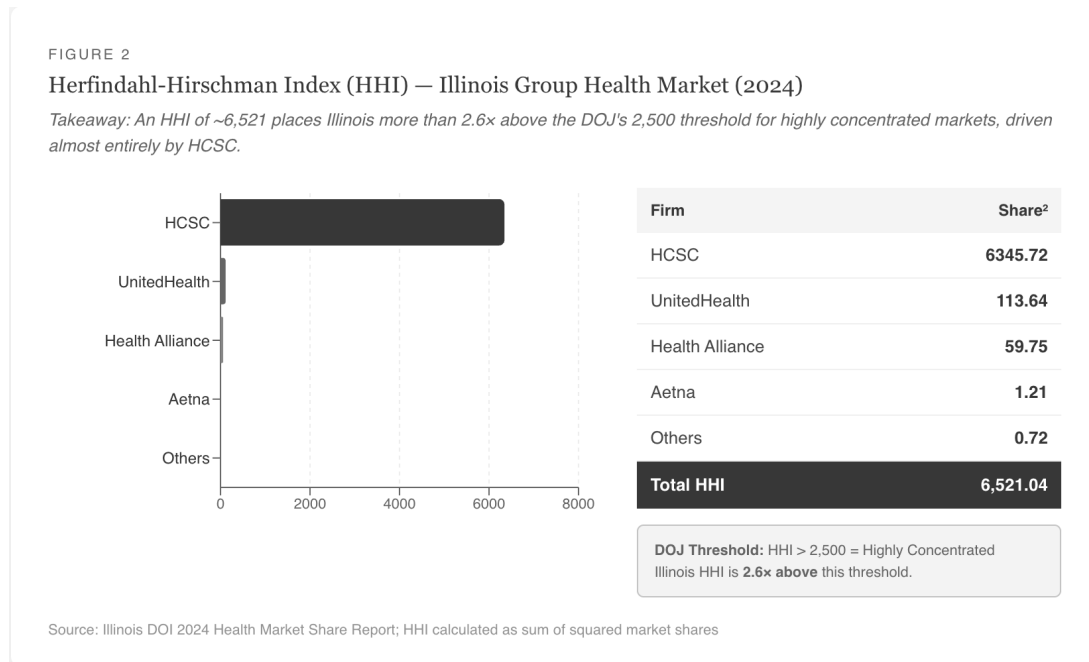


Figure 2: Herfindahl–Hirschman Index decomposition, Illinois group comprehensive health insurance, 2024. The HHI of ~6,521 places Illinois more than 2.6× above the DOJ/FTC’s 2,500 threshold for highly concentrated markets, driven almost entirely by HCSC. Source: author’s calculation from Illinois DOI (2025b); HHI methodology per DOJ and FTC (2023).

The HHI of 6,521 places the Illinois group health insurance market in extreme concentration territory by any standard measure. The U.S. Department of Justice and Federal Trade Commission classify markets with an HHI above 2,500 as highly concentrated, Illinois’s figure is more than 2.6 times that threshold. For context, a perfectly competitive market with many equal-sized firms approaches an HHI near zero, while a pure monopoly produces an HHI of 10,000. At 6,521, Illinois sits closer to monopoly than to competition on that spectrum.

What makes this figure particularly significant from a market organization perspective is the composition of the HHI. In most concentrated markets, a high HHI reflects a small number of large firms each holding meaningful shares, the classic oligopoly. Here, the structure is different: a single firm accounts for 97.3% of the total HHI score. This is not an oligopoly in the traditional sense but a dominant-firm market with a competitive fringe, a structure associated in the market organization literature with distinct strategic behavior. The dominant firm sets the effective price floor and service standard while fringe firms compete for residual demand.

This level of concentration also raises a counterargument worth addressing: high HHI does not automatically imply anticompetitive harm. In some industries, high concentration reflects natural economies of scale, network efficiencies, or superior products rather than exclusionary conduct. The insurance industry has elements of all three, larger insurers negotiate better provider rates, maintain broader networks, and spread administrative costs over more policyholders. HCSC’s dominance may therefore partly reflect genuine efficiency advantages rather than market foreclosure. Whether that efficiency translates to consumer benefit or is retained as insurer profit is precisely what the complaint and loss ratio data in the following tables test.

3.3 Consumer Complaint Evidence

Complaint ratios, measured as complaints per 10,000 policies in force, vary substantially across Illinois's major health insurers (Table 1). HCSC records the lowest ratio at 1.36, followed by Aetna at 1.62, Health Alliance at 3.07, and UnitedHealth at 3.33. Notably, the two firms with the highest complaint ratios, UnitedHealth and Health Alliance, together hold just 18.39% of the market, while the dominant firm, HCSC, despite serving nearly 6 million policyholders, generates the fewest complaints per policyholder of any major insurer.

Table 1: Consumer Complaint Ratios by Insurer, Illinois Health Coverage, 2024.

Firm	Complaints	Policies in Force	Ratio (per 10,000)
HCSC (BCBS)	808	5,960,903	1.36
UnitedHealth	301	903,903	3.33
Aetna	66	408,527	1.62
Health Alliance	36	117,154	3.07

Note. Ratio = (Complaints ÷ Policies in Force) × 10,000. UnitedHealth and Aetna figures aggregate subsidiary entities. Source: Illinois DOI (2025a).

From a consumer welfare standpoint, this finding directly contradicts the baseline prediction of the market power hypothesis. Standard market organization theory predicts that dominant firms, facing weakened competitive discipline, will allow consumer surplus to erode through degraded service quality, manifesting, in this context, as elevated complaint rates as consumers encounter more frequent adverse coverage decisions. Under this logic, HCSC's 79.66% market share should correlate with elevated complaint rates relative to its smaller competitors. The data shows the opposite.

Several mechanisms may explain this apparent paradox. First, the scale itself may confer operational advantages. With nearly 6 million policyholders, HCSC likely achieves greater administrative efficiency, more sophisticated claims processing infrastructure, and deeper provider network relationships than smaller competitors, all of which could reduce the frequency of disputed claims. Second, HCSC's long institutional history in Illinois may reflect accumulated organizational capability that newer or smaller entrants have not yet developed. Third, as discussed in Figure 1, HCSC's dominant position is sustained largely through employer relationships. Employers, unlike individual consumers, are sophisticated repeat buyers with the leverage and incentive to monitor insurer performance and switch providers if service quality deteriorates. This employer-side accountability may impose a competitive discipline on HCSC that individual consumer markets would not.

A critical limitation of complaint ratio data is that it measures reported dissatisfaction, not actual denial rates or coverage outcomes. Consumers who are denied coverage but do not file formal complaints, whether due to lack of awareness, administrative burden, or resignation, are invisible in this data. Complaint ratios, therefore, likely undercount the true frequency of adverse coverage decisions, and may do so unevenly across firms if policyholders of different insurers have different awareness of or access to the complaint process. This limitation is particularly relevant given that the subsequent table shows claims handling, the category most directly tied to denial decisions, drives most complaints across all firms.

3.4 Complaint Composition

Across all four major Illinois health insurers, claims handling is the dominant source of consumer complaints by a substantial margin (Table 2). Claims handling accounts for 79.1% of UnitedHealth's complaints, 75.0% of Health Alliance's, 70.3% of HCSC's, and 53.0% of Aetna's. Underwriting and policyholder service account for the remainder, with marketing and sales representing a negligible share across all firms. In absolute terms, HCSC's 568 claims-handling complaints dwarf those of all other firms combined, though this reflects its dramatically larger policyholder base rather than a disproportionate rate.

Table 2: Complaint Composition by Category, Illinois Health Insurers, 2024.

Firm	Total Complaints	Claims Handling	Underwriting	Policyholder Svc	Claims % of Total
HCSC (BCBS)	808	568	187	106	70.3%
UnitedHealth	301	238	51	28	79.1%
Aetna	66	35	16	16	53.0%
Health Alliance	36	27	7	8	75.0%

Note. A single complaint may have multiple reasons; category counts may exceed total complaints. The marketing/sales category is omitted for brevity (negligible share across all firms). Source: Illinois DOI (2025a).

The consistency of this pattern across firms of vastly different sizes and market positions is the central finding of this paper. Whether a firm controls 80% of the market or 1%, the overwhelming majority of consumer friction originates in claims adjudication, the process by which insurers evaluate, approve, or deny requests for coverage and reimbursement. This industry-wide uniformity has a direct implication for policy: if consumer harm were primarily a function of market power, we would expect complaint composition to differ systematically between dominant and fringe firms. It does not.

From a market organization perspective, this points toward a structural explanation rooted in insurer incentives rather than competitive conduct. Claims adjudication is precisely where insurer and consumer interests most directly conflict. Insurers bear the cost of approved claims and benefit financially from denials, delays, and narrow coverage interpretations. This creates a fundamental incentive, present regardless of market share, to use administrative mechanisms such as prior authorization requirements, step therapy protocols, and medical necessity criteria to manage claim costs. These tools are not anticompetitive in the traditional market organizational sense but represent a form of vertical restraint between insurer and policyholder, embedded in contract terms that most consumers do not read or cannot meaningfully negotiate.

The asymmetric information framework developed in the market organization literature is directly applicable here. Insurers possess far greater information about coverage criteria, denial rates, and appeals outcomes than policyholders do at the time of purchase. This information asymmetry allows insurers to write contracts with restrictive claims adjudication terms that consumers systematically underestimate at enrollment. The result is that adverse coverage decisions, and the complaints they generate, are not primarily a symptom of market concentration but of the inherent information structure of insurance markets, amplified by the employer-intermediated purchasing process in which individual employees have limited ability to evaluate plan quality before enrollment.

This finding does not exonerate market concentration as a concern. A more competitive market might produce greater transparency in claims practices as firms compete on denial rates and appeals outcomes. But the allocative inefficiency this data identifies, the welfare loss from medically necessary care that is denied, delayed, or discouraged through administrative friction, is better understood as a deadweight loss generated by the incentive structure of insurance contracting itself than because of HCSC's market position. The value of approved care to the patient exceeds the cost to the insurer; when that care is withheld through prior authorization denials or step therapy requirements, the surplus that would have been generated is lost entirely rather than transferred. That deadweight loss is present at every insurer in this dataset, regardless of size, which is why complaint composition is uniform across firms spanning 1% to 80% of the market. This does suggest that regulatory interventions focused solely on market structure, such as antitrust enforcement, would be insufficient to address the consumer welfare problem this data identifies. The evidence points toward claims process reform as the more direct lever.

3.5 Loss Ratios

Loss ratios for Illinois group comprehensive health insurers, measured as direct losses incurred divided by direct premiums earned, range from 82.71% for Aetna to 93.05% for Health Alliance (Table 3). HCSC records a loss ratio of 92.24% and UnitedHealth 83.71%. Notably, only two of the four major insurers, HCSC at 92.24% and Health Alliance at 93.05%, exceed the ACA's Medical Loss Ratio floor of 85% for large group plans, while UnitedHealth and Aetna fall marginally below it. This pattern suggests that the dominant firm is not extracting monopoly rents through suppressed loss ratios.

Table 3: Pure Direct Loss Ratios, Illinois Group Comprehensive Health, 2024.

Firm	Loss Ratio	vs. ACA Floor (85%)
Health Alliance	93.05%	+8.05 pp
HCSC (BCBS)	92.24%	+7.24 pp
UnitedHealth	83.71%	-1.29 pp
Aetna	82.71%	-2.29 pp
ACA MLR Floor (Large Group)	85.00%	Regulatory minimum

Note. Loss ratio = Direct Losses Incurred ÷ Direct Premiums Earned. UnitedHealth figure reflects the Illinois-domiciled entity (NAIC 60318). Source: Illinois DOI (2025b); ACA MLR context per KFF (2024).

Loss ratios provide the producer surplus perspective that complaint ratios alone cannot capture. In the standard welfare decomposition, producer surplus represents the share of market value retained by the firm above its costs, in insurance markets, this corresponds roughly to the spread between premiums collected and claims paid, net of administrative costs. A persistently high loss ratio indicates that premium revenue is being converted to claims at a rate that leaves limited margin for insurer profit, while a loss ratio below the regulatory floor indicates that insurers are retaining a larger share of premiums than the ACA's consumer protection threshold permits. From a consumer welfare standpoint, higher loss ratios are generally favorable, they suggest premiums are functioning as intended, converting policyholder contributions into medical care rather than insurer margin.

The variation in loss ratio compliance across firms is itself informative. HCSC and Health Alliance exceed the 85% MLR floor by 7.24 and 8.05 percentage points, respectively, while UnitedHealth and Aetna fall modestly below it by 1.29 and 2.29 percentage points. This pattern partially addresses the concern raised by the HHI analysis: despite extreme market concentration, HCSC is not extracting monopoly rents through suppressed loss ratios. In fact, HCSC pays out more per premium dollar than either of its nationally operating competitors, both of which function in more competitive markets yet retain a slightly larger share of premiums above the regulatory limit. This suggests the ACA's MLR floor may be functioning as a binding constraint on the fringe competitors in this market more than on the dominant firm, an unexpected finding that warrants attention.

This, however, raises its own counterargument. Loss ratios measure the aggregate ratio of claims paid to premiums collected, but they do not capture which claims are paid and which are denied. An insurer could maintain a high loss ratio while still systematically denying high-cost surgical procedures or specialty care if the volume of approved routine claims is sufficiently large. The loss ratio is therefore a necessary but not sufficient indicator of consumer welfare, it tells us that money is flowing back to policyholders in aggregate, but not whether the distribution of that spending reflects medically appropriate coverage decisions.

Furthermore, the variation in loss ratios across firms, nearly 11 percentage points between Aetna at 82.71% and Health Alliance at 93.05%, warrants attention. Aetna's relatively lower loss ratio, combined with its small market share and moderate complaint ratio, may reflect a more selective enrollment strategy targeting lower-risk employer groups, or more aggressive claims management practices, or both. Disentangling these explanations would require claims-level data beyond what is publicly available from the Illinois DOI, representing a meaningful limitation of this analysis.

Taken together, the loss ratio evidence suggests that the ACA's MLR provision is functioning as an effective regulatory constraint on insurer profitability in the Illinois group market, even under conditions of extreme concentration. The more pressing efficiency concern, as the complaint composition data makes clear, lies not in the aggregate flow of premiums to claims but in the adjudication process that determines which claims are approved.

4 CONCLUSION

This paper set out to examine whether high market concentration in the Illinois group health insurance market produces measurable consumer harm. The evidence assembled from Illinois Department of Insurance data for 2024 tells a consistent but counterintuitive story: extreme concentration, as measured by an HHI of approximately 6,521, does not manifest in elevated complaint rates against the dominant firm. HCSC, which controls nearly 80% of the group comprehensive market, generates fewer complaints per policyholder than any of its significantly smaller competitors. The welfare concerns this market presents are real, but their source is not where standard market power theory would direct our attention.

The three principal findings of this paper point toward a different diagnosis. First, HCSC's dominance appears rooted in employer demand-side preferences, network breadth, administrative

reliability, and institutional familiarity, rather than anticompetitive foreclosure of rivals. In the employer-sponsored insurance market, employers function as sophisticated repeat buyers who impose a form of competitive discipline on dominant insurers that individual consumer markets would not generate. This demand-side explanation for concentration has important implications: it suggests that simply reducing HCSC's market share through antitrust intervention would not necessarily improve the consumer experience if the underlying structural features of the employer-sponsored market remain intact.

Second, and most significantly, claims handling accounts for most consumer complaints across every insurer in the Illinois market, regardless of size or market position. This industry-wide uniformity points to the incentive structure of insurance contracting itself as the primary driver of consumer friction. All insurers, dominant and fringe alike, face the same fundamental tension between approving claims that benefit policyholders and managing costs that affect profitability. Prior authorization requirements, step therapy protocols, and medical necessity criteria are the administrative instruments through which this tension is resolved, and they generate disputes regardless of whether the insurer controls 80% or 1% of the market. The asymmetric information embedded in insurance contracts, combined with the moral hazard management pressures all insurers face, creates a structural predisposition toward claims friction that market structure alone cannot explain or remedy.

Third, loss ratio evidence presents a nuanced picture of regulatory effectiveness. HCSC and Health Alliance exceed the ACA's 85% MLR floor for large group plans, while UnitedHealth and Aetna fall marginally below it. Counterintuitively, the dominant firm is paying out the second-highest share of premiums as claims, suggesting that extreme concentration has not translated into the kind of profit extraction the MLR rule was designed to prevent, at least as measured by this metric.

Taken together, these findings support a reorientation of the policy conversation around the Illinois health insurance market. The dominant policy concern should not be market concentration per se, but the claims adjudication practices that generate consumer harm across all insurers uniformly. The interventions most directly responsive to the evidence assembled here would target transparency and accountability in the claims process, requiring public disclosure of prior authorization denial rates by service category, strengthening external appeals rights for denied claims, and imposing penalties when initially denied claims are systematically overturned on appeal. The KFF data on Medicare Advantage prior authorization, while not directly applicable to the commercial market analyzed here, illustrates the scale of the problem: in 2024, Medicare Advantage insurers denied 7.7% of prior authorization requests, and of the small share that were appealed, over 80% were overturned (KFF, 2026), suggesting a pattern of initial over-denial that a more accountable process could correct.

This paper has several limitations that future research should address. The cross-sectional design prevents causal inference about the relationship between market structure and outcomes. Complaint ratios undercount adverse coverage decisions by an unknown and likely substantial margin. Illinois-specific commercial prior authorization data is not publicly available, leaving the most direct measure of claims adjudication behavior unobserved. And the analysis covers only one year, making it impossible to assess whether current patterns reflect stable structural features or transient conditions.

Future research should pursue claims-level denial data from state insurance regulators, leverage natural experiments created by insurer entry and exit to identify causal effects of concentration on consumer outcomes, and examine whether the employer-intermediated purchasing structure systematically suppresses complaint filing relative to individual markets. The Illinois market, precisely because of its extreme concentration and rich regulatory data environment, offers an unusually clean setting for such investigations.

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